



Distinctive patterns of Multiple Object-Tracking performance trajectories in youth with deficits in attention, learning, and intelligence

Domenico Tullo^{a,*}, Jocelyn Faubert^b, Armando Bertone^c

^a School of Education, University of California, Irvine, 3200 Education Bldg, Irvine, CA 92697, USA

^b École d'optométrie, University of Montreal, 3744, rue Jean-Brillant, Montréal, QC H3T 1P1, Canada

^c Department of Education & Counselling Psychology, McGill University, 3700 McTavish Street, Montréal, H3A 1Y2 QC, Canada

ARTICLE INFO

Keywords:

Attention

Learning

Neurodevelopmental conditions

Multiple object-tracking

ABSTRACT

There is a significant overlap in symptomology between individuals with deficits in attention and learning, which is explained by the co-dependent dynamic between the two cognitive constructs. Within this dynamic, attentional resources are allocated to salient stimuli while learning mechanisms distinguish relevant from irrelevant information. Moreover, individuals with deficits in higher-order cognition (i.e., intelligence) can demonstrate difficulties in attention and learning. The Multiple Object-Tracking (MOT) task is a sensitive and versatile measure of attention that has characterized individual differences in attention as a function of higher-order cognition. Exploiting the traditional MOT task's ability to characterize the allocation of attentional resources to task demands, the current study compared learning exhibited on an attention-based task across neurodevelopmental conditions defined by deficits in attention (attention-deficit/hyperactivity disorder; ADHD), learning (specific learning disorder; SLD), and intelligence (intellectual developmental disorder; IDD). Children and adolescents ($N = 101$) completed 15 sessions on a Multiple Object-Tracking (MOT) task where performance trajectories were analyzed using latent growth curve modeling and conditioned by the presence of ADHD, SLD, or IDD while controlling for performance on a separate measure of attention, age, and sex. The sample, characterized by below-average IQ and problematic levels of attention, exhibited an effect of learning on MOT. However, individuals with an IDD diagnosis demonstrated decreased baseline MOT capability while ADHD and SLD profiles exhibited decreased slopes, relative to other neurodevelopmental conditions. Taken together, the results demonstrate distinct linear performance trajectories between neurodevelopmental conditions defined by deficits in attention, learning, and intelligence. The current study provides additional evidence to repurpose the traditional MOT task as a descriptor of attention and discusses alternative uses for the paradigm. Overall, these results suggest an eclectic approach that considers attention, learning, and higher-order cognition when diagnosing ADHD, SLD, or IDD.

Abbreviations: MOT, Multiple Object-Tracking; ADHD, Attention Deficit/Hyperactivity Disorder; SLD, Specific Learning Disorder; IDD, Intellectual Developmental Delay.

* Correspondence to: School of Education, University of California, Irvine, 3200 Education Bldg, Irvine, CA 92697, USA.

E-mail address: dtullo@uci.edu (D. Tullo).

<https://doi.org/10.1016/j.ridd.2022.104402>

Received 24 August 2021; Received in revised form 23 November 2022; Accepted 16 December 2022

Available online 22 December 2022

0891-4222/© 2022 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

What this paper adds

There is a significant overlap in presenting symptoms between individuals diagnosed with attention-deficit/hyperactivity disorder (ADHD) and specific learning disorder (SLD), which is explained by the co-dependent relationship between attention and learning. The current study compares learning capability on an attention-based task to isolate condition-specific deficits across ADHD, SLD, and intellectual developmental disorder (IDD) diagnostic profiles. Specifically, we examined differences in performance trajectories in Multiple Object-Tracking (MOT) across neurodevelopmental conditions defined by deficits in attention, learning, and intelligence. To examine repeated practice on this measure of attention, we employed latent growth curve modeling to characterize performance, which allowed us to examine the differences in baseline as well as the rate of change in MOT performance across the neurodevelopmental conditions. Here, we highlight the role of attentional capacity and learning capability to characterize the difficulties expressed by individuals with ADHD, SLD, and IDD. Overall, these results have both theoretical and clinical implications. For instance, the findings provide support for an alternative avenue to measure, analyze, and interpret learning exhibited on an attention-based paradigm. Moreover, these results suggest that practitioners should consider deficits in attention, learning, and higher-order cognition when diagnosing or characterizing capability for individuals with ADHD, SLD, and IDD.

1. Introduction

The robust and clinically relevant relationship between attention and learning (Mayes et al., 2000) can be examined by the interaction between the two cognitive constructs. More specifically, the interplay between attention and learning can be represented by repeated practice on an attention-based task (Seitz & Watanabe, 2005, 2009). Improved performance requires orienting attentional resources to task information, which is reinforced by the individual's ability to distinguish between relevant and irrelevant information throughout repeated practice (Seitz & Watanabe, 2005). The relationship between attention and learning is therefore characterized as a co-dependent dynamic where (i) attentional resources focus on target and distractor properties, and (ii) learning mechanisms, refine the efficiency of directing attentional resources towards relevant, task-specific information while suppressing the allocation of resources to irrelevant, distracting information (Kelley & Yantis, 2009). This reciprocal relationship illustrates the mechanisms underlying task mastery where attentional resources are spared from irrelevant information, increasing the likelihood of successfully completing the task at higher levels of difficulty (Sweller, 1988; Zeithamova & Maddox, 2006).

The mechanisms underlying this reciprocal relationship are exhibited in everyday activities. Successfully completing attention-based tasks, such as finding a salient object within a crowded scene, tracking the position of teammates and opponents while playing a sport, or attending to relevant classroom material while ignoring distracting events, depend on the deployment and availability of sufficient attentional resources to satisfy the task's demands (Ennesser & Medioni, 1995; Faubert & Sidebottom, 2012; Steinmayr et al., 2010). Conversely, failed attempts can be the result of either incorrectly deploying resources to irrelevant information (Doshier et al., 2010; Sweller, 1988, 1994) or a deficiency of attentional resources to satisfy task demands (Alvarez & Franconeri, 2007). Available attentional resources are pooled within a finite cognitive reserve referred to as attention resource capacity (Alvarez & Franconeri, 2007), which varies in size across the general population (Alnæs et al., 2014; Tullo, Faubert et al., 2018). Specifically, individuals with large capacities are better at allocating and maintaining sufficient resources to task demands compared to individuals with smaller capacities (Kane et al., 2007). In fact, attention resource capacity has been demonstrated to be related to successful task completion and learning capability (Engle, 2002; Engle et al., 1999; Gathercole & Alloway, 2008). Moreover, individual differences in capacity are accounted for by chronological age (Klingberg, 2013) as well as intelligence (Engle et al., 1999; Tullo, Faubert et al., 2018), and these differences are highly prominent between typically and atypically developing populations (Case, 1985; Melby-Lervåg et al., 2016).

Individuals with deficits in attention, learning, or intelligence are vulnerable to distracting and/or irrelevant information, which impedes task completion because of the limited availability of attentional resources for task demands (Deutsch et al., 2008; Forster & Lavie, 2008; Forster et al., 2014; Passolunghi et al., 2005). Attention-deficit/hyperactivity disorder (ADHD) and specific learning disorder (SLD) are two of the most prevalent neurodevelopmental conditions, and often co-occur (Zablotsky et al., 2019). The association between these neurodevelopmental conditions can be in part explained by the overlap in symptomology where (i) difficulties in attention are exhibited in some individuals diagnosed with SLD, but not ADHD, and (ii) difficulties in learning are presented in some individuals with ADHD that do not have a concurrent SLD diagnosis (Mayes et al., 2000; Pondé et al., 2011). Additionally, attention and learning difficulties can also present in individuals with intellectual developmental disorder (IDD; Baker et al., 2010; Wilson et al., 2013), where the overlap in symptomatology can result from atypically developing higher-order cognition (Deutsch et al., 2008).

The overlap in symptomology between the aforementioned three neurodevelopmental conditions has demonstrated the need to characterize the attention-learning dynamic for individuals with and without deficits in attention, learning, and/or intelligence (Soo & Bailey, 2006). Although many of the measures and approaches used to assess attention isolate the specific underlying subcomponents (Kelley & Yantis, 2009; Seitz & Watanabe, 2005), a more representative measure of attentional ability is attention resource capacity, an eclectic metric comprised of multiple attentional subcomponents (Scholl, 2009). Specifically, selectively assigning, sustaining, and distributing attentional resources between relevant information while suppressing the allocation of attentional resources to irrelevant information is a significant predictor of both successful task completion and learning ability (Powell et al., 2017; Scerif, 2010). Moreover, examining performance trajectories across repeated practice sessions can characterize learning capability on the attention-based task.

The MOT paradigm is a validated and traditional psychophysical measure of multiple subcomponents of attention (i.e., sustained, selective, and distributed) that is ideal for describing the attention-learning dynamic across deficits in attention, learning, and intelligence. The task involves tracking a subset of target items while ignoring physically indistinguishable distractor items over a short

period of time and within a limited space. Designed by researchers within the field of vision sciences (Pylyshyn & Storm, 1988), MOT has been used to define the limits of visual cognition (Scholl, 2009). However, an alternative area of research that is currently developing has repurposed MOT as a robust measure of attention (Huang et al., 2012; Scholl, 2009; Tullo et al., 2020), demonstrating the task's sensitivity to differences across periods of development (Brockhoff et al., 2016; Trick, Jaspers-Fayer et al., 2005) and between typically- and atypically-developing individuals (Evers et al., 2014; O'Hearn et al., 2010; Van der Hallen et al., 2015). In fact, previous research has characterized individual differences in attention resource capacity, defined by MOT task performance, as a function of higher-order cognition (Oksama & Hyönä, 2004; Tullo, Faubert et al., 2018). Moreover, research by leaders in the field of Multiple Object-Tracking has characterized the task as purely attentional in nature (Meyerhoff & Papenmeier, 2020; Scholl, 2009). In fact, previous research characterizing brain-related activations (i.e., using EEG) during the tracking phase demonstrated specific activations to brain regions that are distinctly linked to attention (see Parsons et al., 2014; Parsons & Faubert, 2021). Therefore, examining performance trajectories, as defined by repeated practice on the MOT task, can characterize the allocation and maintenance of attentional resources to task demands (i.e., attention resource capacity).

The current study would be the first to define and contrast attention and learning capabilities for individuals with ADHD, SLD, and IDD diagnostic profiles by assessing performance trajectories after repeated practice on a MOT task. Although other studies have examined learning an attention-based task via repeated MOT sessions (Makovski & Jiang, 2009; Strong & Alvarez, 2017), the methodological approaches used did not examine MOT performance as a measure of attention. For example, Makovski and Jiang (2009) standardized the trajectories for MOT distractor objects and therefore, examined implicit memory processes rather than the allocation of attentional resources. Similarly, Strong and Alvarez (2017) manipulated aspects of visual cognition within the MOT task to examine whether learning MOT transferred across other conditions that can be characterized as sensory dependent. For example, Strong and Alvarez (2017) examined transfer in tracking capability across sensory dependent conditions of motion and location. In addition, repeated practice on MOT has translated to improvement on separate measures of attention and higher-order cognition in a sample of neurotypical adults (Parsons et al., 2014; Vartanian et al., 2016) and youth with diverse neurodevelopmental conditions (Tullo & Guy, 2018). Capitalizing on the MOT task's accuracy in isolating attention resource capacity, Tullo, Guy, et al. (2018) randomized control trial intervention study demonstrated that the benefits of repeated practice on MOT transferred to improvement on a clinically validated measure of attention in youth diagnosed with a neurodevelopmental condition. However, it remains unknown how the improvement within the attention-based visual task differed between neurodevelopmental conditions.

The purpose of this study is to characterize learning an attention-based task for individuals with and without deficits in attention, learning, and intelligence by assessing differences in performance trajectories between individuals diagnosed with a neurodevelopmental condition. To achieve this objective, we analyze performance trajectories across 15 MOT sessions using latent growth curve modeling (LGCM), conditioned as a function of diagnostic predictors (i.e., the presence of ADHD, SLD, and/or IDD). We expect fitted performance trajectories to suggest an overall increase in tracking capability, further demonstrating that MOT is accessible and learnable for individuals with different cognitive abilities, as demonstrated by previous research (Faubert, 2013; Makovski & Jiang, 2009; Tullo & Guy, 2018). Given the overlap between ADHD and SLD symptomatology (Mayes et al., 2000) and the attention-learning dynamic (Kelley & Yantis, 2009), we hypothesize reduced learning in MOT performance for individuals that are diagnosed with the presence of attention- or learning-based deficits compared to all other diagnostic profiles. Additionally, we expect an overall reduced MOT performance for participants with deficits in intelligence compared to all diagnostic profiles, given previous findings that MOT performance is related to intelligence (Tullo, Faubert et al., 2018).

2. Materials and method

The Research Ethics Board (REB) at *McGill University* has approved the protocol for this project. Informed consent was obtained from all participants and/or their legal guardians prior to experimentation. All methods were performed in accordance with the relevant guidelines and regulations approved by the REB at *McGill University*. The datasets generated and/or analyzed during the current study are not publicly available because the data contains potential identifying info from atypically developing children recruited from a research center that upholds full confidentiality.

2.1. Participants

Participants between 6 and 17 years of age were recruited from three schools in Québec, Canada, that provide specialized services for students with alternative learning profiles. A consent form was sent home to all students enrolled at three participating schools between the September and December months in the academic year in which the study took place at that school. Participants consisted of all individuals that consented to participate within the study's timeline.

One-hundred and six participants' parents provided consent to participate in the study and children gave assent; however, 5 individuals were excluded as their diagnostic status was either unconfirmed or pending at the time of statistical analysis. The final 101 participants included in the study ($M_{\text{age}} = 13.58$, $SD_{\text{age}} = 1.83$) previously received a diagnosis of a neurodevelopmental condition by a clinician and when possible was verified by either (i) codes established by Ministère de l'Éducation de l'Enseignement Supérieur (MEES or the Quebec Ministry of Education and Higher Education), or (ii) confirmatory assessments provided by licensed school psychologists at each site. For example, the MEES sets out specific criteria and codes to identify students with neurodevelopmental conditions; such as SLD and IDD, and educational needs (Fombonne et al., 2006). All participants receiving a code would have been evaluated by an expert clinician working with a multidisciplinary or interdisciplinary team (Government of Québec, 2013). Furthermore, all participants had normal or corrected to normal visual acuity and stereopsis at the time of testing, confirmed using Directional 'E' and 'C' near

visual acuity charts, and the RANDOT Stereotest, respectively. All participants successfully completed both tests and thus, no participants were excluded from the study for reasons related to vision.

We created three binary variables representing the three diagnostic profiles of interest, where the presence of that neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the reference category (i.e., 'Yes'). The absence of a diagnosis for that neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the non-reference category (i.e., 'No'; see Table 1). For example, participants with a confirmed diagnosis of ADHD (i.e., primary or secondary) were assigned to the reference category (i.e., 'Yes') for the ADHD predictor variable whereas participants with SLD, IDD, or any other neurodevelopmental condition except for ADHD was listed in the non-reference category for the same ADHD predictor variable. This approach was undertaken to account for comorbidity typically expressed by individuals diagnosed with ADHD, SLD, and IDD. Furthermore, the dichotomization of diagnostic status loosened the exclusionary criteria and allowed for an evaluation of a sample that best reflects populations typically underserved and underrepresented in this area of research. The frequency of diagnostic status and comorbidities across the three neurodevelopmental conditions of interest is presented in Table 2.

2.2. Measures

2.2.1. Three-Dimensional Multiple Object-Tracking (MOT)

Learning an attention-based task was examined by repeated practice on the same variant of the MOT task used in our group's previous work that (i) characterized individual differences in attentional capacity across neurotypical adults (Tullo, Faubert et al., 2018); (ii) investigated learning MOT across levels of athleticism in adults (Faubert, 2013), and (iii) evaluated the efficacy of MOT training to transfer to a separate, clinically validated measure of attention (Parsons et al., 2014; Tullo & Guy, 2018). The MOT task was presented on a 51" (i.e., 1.3 m) active 3D-TV. Participants sat in a chair that was positioned approximately 1.52 m away from the screen and wore LCD shutter 3D glasses.

A MOT trial is comprised of five segments: (i) presentation, (ii) identification, (iii) tracking, (iv) response, and (v) feedback (Fig. 1). The trial began with a two-second presentation of all eight physically indistinguishable objects (i.e., yellow spheres; Fig. 1A). Next, three target objects were indexed for three seconds by changing from yellow to orange (Fig. 1B). All target items returned to their original colour and started moving for a period of 8s, participants were asked to track all three target items previously identified in the second stage (Fig. 1C). Participants were informed that they had to stare at the fixation point throughout the tracking phase. After the tracking stage, all items stop moving and are labeled from 1 to 8 (Fig. 1D). Participants reported their responses to the trial by selecting target items using a numerical keypad. There was no time limit for participants to report their responses. Upon confirming their responses, feedback was provided to the participant for three seconds, which confirmed whether they selected the correct target items or instead highlighted the correct target items (see Fig. 1E). The trial restarted after the presentation of feedback. The task adapted to the participants' capability as a function of object velocity in the subsequent trial. Specifically, the speed of all objects in the tracking stage increased after a correct response (i.e., identifying *all* target items) and decreased after an incorrect response using a 1 up 1 down staircase procedure. The first trial was set at 68 cm/s and item speed increased or decreased by 0.05 log units depending on the outcome of the previous trial (see Levitt, 2005 for details on staircase procedure). The range in speed for object velocity was from 0.68 cm/s to 544 cm/s. Performance was defined as the average speed score, calculated as the geometric mean of speed at each inversion (i.e., a shift from a correct to an incorrect answer and vice versa), across six total inversions (see Legault & Faubert, 2012; Tullo, Faubert et al., 2018; for more details on the task's psychometrics).

Table 1
Participant characteristics by diagnostic profile.

Measure	ADHD			IDD			SLD			Total
	Yes	No	<i>p</i> (<i>d</i>)	Yes	No	<i>p</i> (<i>d</i>)	Yes	No	<i>p</i> (<i>d</i>)	-
<i>n</i>	47	54		32	69		14	87		101
Male:Female	30:17	36:18		21:11	45:24		7:7	59:28		66:35
Age	13.64 (1.14)	13.50 (2.26)	.702 (.08)	14.32 (1.41)	13.22 (1.89)	*.004 (.59)	13.61 (0.82)	13.56 (1.94)	.850 (.04)	13.56 (1.82)
FSIQ	78.34 (12.84)	75.13 (14.27)	.240 (.24)	66.00 (8.66)	81.55 (12.74)	*<.001 (1.26)	83.29 (12.09)	75.55 (13.56)	.049 (.40)	76.62 (13.65)
PRI	83.45 (15.30)	78.13 (17.75)	.113 (0.32)	67.84 (10.32)	86.52 (15.93)	*<.001 (1.42)	86.71 (11.26)	79.62 (17.36)	.143 (.30)	80.60 (16.79)
VIQ	77.19 (12.55)	75.83 (13.98)	.611 (.10)	68.75 (10.33)	80.04 (13.03)	*<.001 (.94)	83.21 (13.45)	75.38 (13.01)	.040 (0.42)	76.47 (13.28)
<i>d'</i> t-score	60.72 (7.49)	59.13 (9.77)	.365 (.18)	58.69 (9.62)	60.42 (8.37)	.358 (.19)	61.86 (8.33)	59.55 (8.77)	.364 (.18)	59.87 (8.78)

Means and (standard deviations) are reported for age, WASI-II Full-Scale IQ (FSIQ), WASI-II Perceptual Reasoning Index (PRI; non-verbal IQ), WASI-II Verbal Intelligence Quotient (VIQ), and CPT-3 *d'* t-score. The *p*-value and (Cohen's *d* effect size) is reported for each preliminary independent sample *t*-test analysis conducted between the presence and absence of the neurodevelopmental condition. *Denotes statistically detectable difference at Bonferroni corrected alpha level (.003). Diagnostic predictor variables were divided between the presence of that neurodevelopmental condition (i.e., ADHD, SLD, or IDD), represented as the reference category (i.e., 'Yes'); and the absence of a diagnosis of the respective neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the non-reference category (i.e., 'No').

Table 2
Frequency of diagnostic profiles across conditions of interest.

ADHD	IDD	SLD	Count
YES	YES	YES	2
YES	YES	NO	10
YES	NO	YES	10
YES	NO	NO	25
NO	YES	YES	1
NO	YES	NO	19
NO	NO	YES	1
NO	NO	NO	33

A 'Yes' indicates the presence of the neurodevelopmental condition whereas a 'No' suggests that the absence of that neurodevelopmental condition.

2.2.2. Conners Continuous Performance Task – 3rd Edition (CPT-3)

The CPT-3 is a computer-based, clinically validated measure of attention. Participants are tasked with identifying targets (i.e., any letter of the alphabet except for the letter 'x') by pressing the spacebar when they flash on the computer screen as quickly and as accurately as possible and inhibit this behaviour for non-target items (i.e., the letter 'x'). The 14-minute assessment consists of 360 trials and performance is defined as the participant's ability to detect targets from non-targets: d' . Moreover, the CPT-3 provides age- and sex-based norms on a t -distribution characterizing inattention, impulsivity, and sustained attention. Used in clinical settings as one measure to suggest deficits in attention, the d' t -score characterizes attention as: problematic (i.e., t -score greater than 60); below-average (i.e., t -score between 55 and 59; average performance); average (i.e., t -score between 45 and 55); and above-average (i.e., t -score below 45; [Conners, 2014](#)). Please refer to the Conners CPT-3 manual for detailed information regarding the calculation of the d' metric and the age and sex normed t -score ([Conners, 2014](#)).

2.2.3. Wechsler Abbreviated Scale of Intelligence – 2nd Edition WASI-II

The WASI-II is an abbreviated measure of intelligence (IQ) for individuals between the ages of 6–89 consisting of 4 subtests; block design, matrix reasoning, vocabulary, and similarities ([Wechsler, 2011](#)). From these, Verbal and Performance (non-verbal) IQ index scores are calculated, which together provide a general estimate of intelligence, or Full-Scale IQ (FSIQ). THE WASI-II takes approximately 45–60 min to administer.

2.3. Procedure

All assessments of IQ (WASI-II) and attention (CPT-3), as well as repeated practice on MOT, were conducted in a quiet room located in the participants' school. First, all participants were assessed on the WASI-II and CPT-3 on separate days before beginning MOT sessions.

2.3.1. MOT repeated practice

Prior to starting the first MOT learning session, we assessed the participant's understanding of the task and their capability to complete the task. A trained research assistant dictated the instructions to the participant. To qualify for the study, participants were asked to follow one of eight spheres for 8 s to demonstrate competence in two sets of three trials. If the participant successfully obtained a correct trial (i.e., successfully tracked one target object of eight) in two out of the three first trials, they moved onto the first session. If the participant failed to identify the target object in the first set of three trials, instructions were re-read to the participant, and they were given another set of three trials. Failing to complete the second set of trials would have resulted in exclusion from the study. All 101 participants qualified for the study and completed 15 sessions of MOT, 3 times a week over a period of 5 weeks (i.e., one session per calendar day). Each session consists of two sets of trials (i.e., two average speed scores generated from six inversions). Performance per session was defined as the arithmetic mean between the two average speed scores. A session lasted approximately 7 min in duration.

2.4. Analysis

Prior to the main analysis, we conducted preliminary analyses to examine any pre-existing effects of development, intelligence, and attention. A total of nine independent sample t -tests were conducted to examine whether the presence or absence of the three neurodevelopmental conditions (i.e., ADHD, SLD, IDD) differed across age, intelligence (i.e., WASI-II FSIQ, PRI; and VCI), and CPT-3 (i.e., d' t -score) performance. Furthermore, we applied a Bonferroni corrected alpha value of 0.003 to account for multiple comparisons.

Analyzing the repeated practice on the MOT task using latent growth curve modeling LGCM can characterize the nature of learning trajectories (i.e., linear versus quadratic trends) and examine factors that account for variability in baseline (i.e., intercept) and rate of change (i.e., slope) for MOT performance. We used LGCM with structural equation modeling (LGCM) to examine whether diagnostic profiles, defined by deficits in attention, learning, and intelligence predicted performance on repeated practice of MOT while controlling for (i) CPT-3 d' t -score performance as a separate, clinically validated measure of attention, (ii) age, and (iii) sex. The LGCM analyses were conducted using *R studio* 1.2.5019, with the *lavaan* package version 0.6–5 ([Rosseel, 2012](#)). The R code for the LGCM

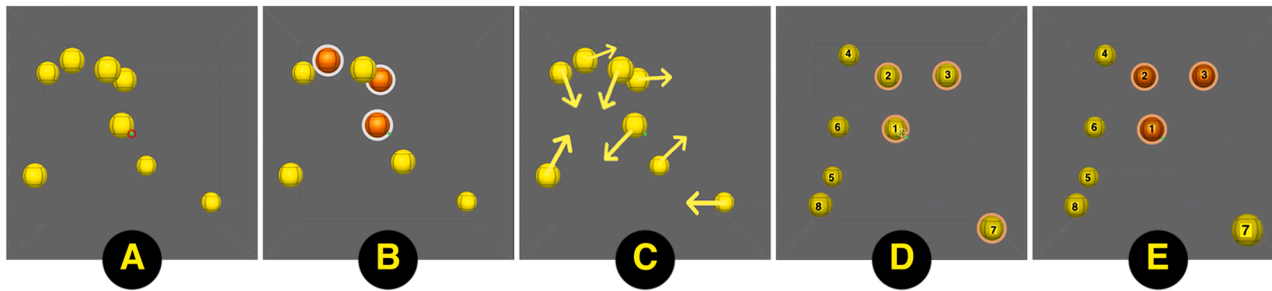


Fig. 1. An illustration of a single trial in the Multiple Object-Tracking task. A trial is comprised of 5 stages: (A) All 8 items (i.e., spheres) were presented for two seconds. Participants were informed that they had to stare at the fixation point throughout the tracking phase (i.e., a red circle is displayed to illustrate the fixation point). (B) Target items are highlighted in yellow for three seconds and to be maintained in tracking; (C) Items move linearly and randomly for 8 s; (D) numbers appear on all 8 items and the participant has no time limit to identify the originally highlighted items; (E) feedback is provided for the participant, where correct items are highlighted for three seconds.

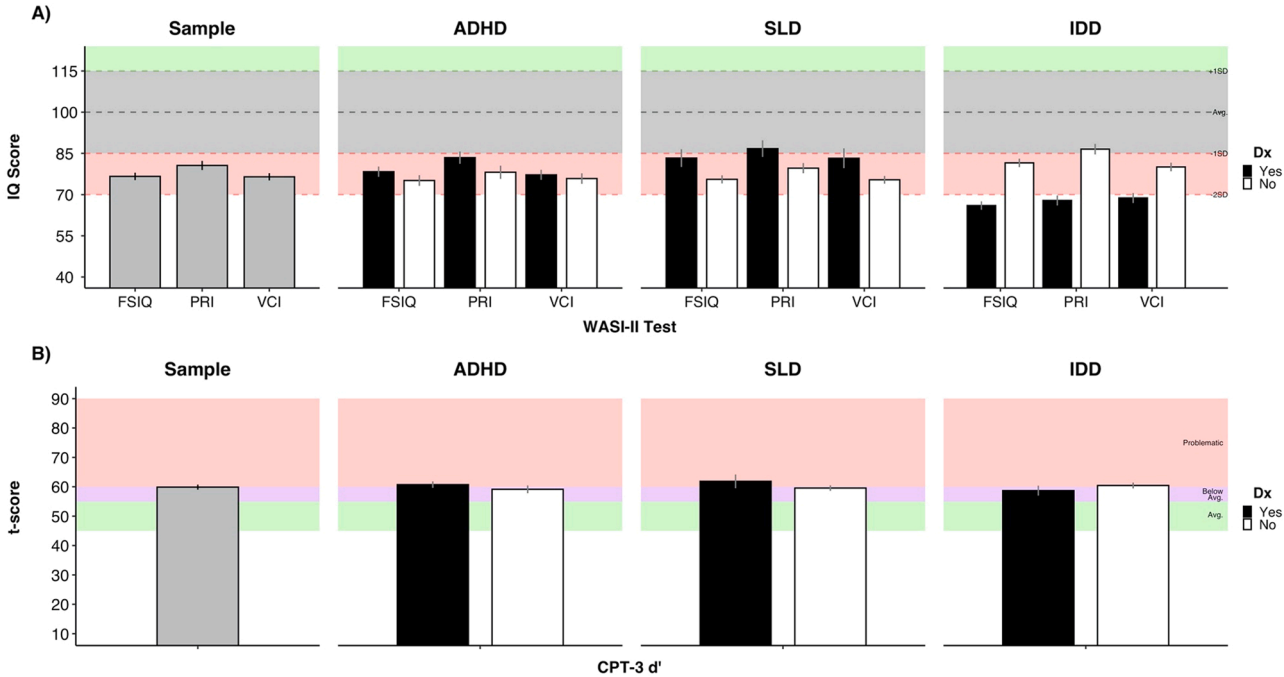


Fig. 2. IQ and CPT-3 performance by dichotomous diagnostic predictor variables (A) Performance on the WASI-II measure of Intelligence. (B) Performance on the CPT-3. Note. The standard error for each group is illustrated by the error bars. Diagnostic predictor variables were divided between the presence of that neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the reference category (i.e., 'Yes'); and the absence of a diagnosis of the respective neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the non-reference category (i.e., 'No').

analysis was written using [Bauer and Curran \(2019\)](#) guide. To examine the learning trajectories across the three different neurodevelopmental conditions using LGCM, we first evaluated the data's fit on an intercept-only model to unconditional linear and quadratic models before testing the hypothesized conditional model with diagnostic variables and measure of attention with the Maximum Likelihood Robust estimator applied ([Bauer & Curran, 2019](#)). There was no missing data for diagnostic predictor variables (i.e., ADHD, SLD, and IDD), nor preliminary measures of age, WASI-II, and CPT-3. We note 0.79% of total missing MOT data, where missing data per participant ranged from missing 1 session (n = 4) to missing 4 sessions (n = 1). All missing values were estimated using full information maximum likelihood.

3. Results

3.1. Preliminary analyses

The means and standard deviations for participant demographics as well as the results of the preliminary analyses are presented in [Table 1](#). There was no statistically detectable difference in age between those with and without an ADHD or SLD diagnosis. As expected, the average cognitive ability for the sample recruited for this study, defined by Weschler IQ score, fell between one and two standard deviations below the population average, which is consistent with below-average to borderline levels of intelligence ([Fig. 2A](#)). This is not surprising since the majority of students diagnosed with a neurodevelopmental condition who receive specialized services often have a below-average cognitive ability ([Boyle et al., 2011](#)). Additionally, there was a statistically detectable difference in age between those with and without an IDD diagnosis. Moreover, there were no statistically detectable differences in WASI-II FSIQ scores between those with and without an ADHD and SLD diagnosis. There was also a statistically detectable difference in FSIQ scores between participants with and without IDD, which was expected given the primary concerns of the neurodevelopmental condition.

Lastly, we found no statistically detectable differences in CPT-3 d' *t*-score performance within neurodevelopmental predictors. As demonstrated in [Fig. 2B](#), the sample's average performance on the CPT-3 ranged from below-average to problematic levels, suggestive of a deficit in attention. Given (i) the sample's collective performance at problematic and/or below-average levels of attention and (ii) the expression of deficits in attention across the three diagnostic profiles of interest, CPT-3 performance was included as a covariate in the proceeding models to account for the homogeneity of the sample.

3.2. Latent growth curve models

We examined the appropriateness of four LGCMs building from the intercept-only model to the hypothesized conditional model (see [Bauer & Curran, 2019](#) for tutorial). Our results demonstrated poor fit indices for the intercept-only model, while the unconditional

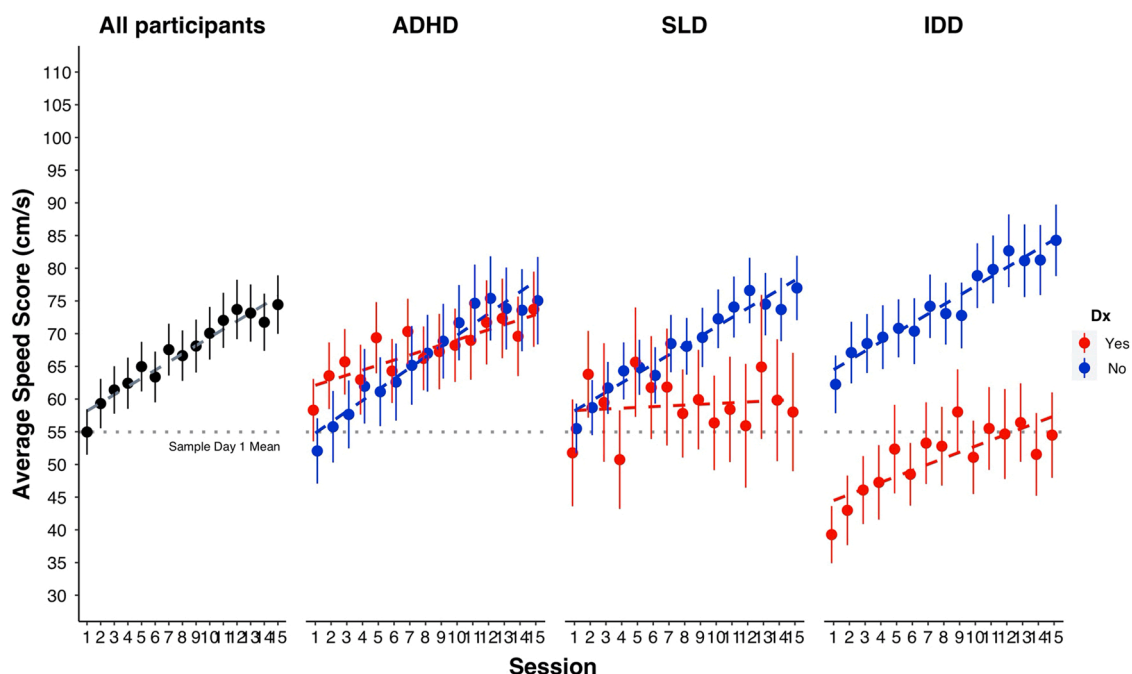


Fig. 3. Learning MOT by dichotomous diagnostic predictor variables. Attention Deficit/Hyperactivity Disorder: ADHD; Specific Learning Disorder; SLD, and Intellectual Development Disorder; IDD). Diagnostic predictor variables were divided between the presence of that neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the reference category (i.e., 'Yes'); and the absence of a diagnosis of the respective neurodevelopmental condition (i.e., ADHD, SLD, or IDD) was represented as the non-reference category (i.e., 'No').

linear model demonstrated excellent fit indices, and the likelihood ratio test comparing the value-added of the linear slope as a latent variable was statistically detectable: $\Delta\chi^2(3) = 101.55, p < .001$, indicating a more suitable model. Furthermore, we tested the value-added of a latent variable assessing quadratic growth given previous findings (see [supplemental table 1](#) for fit indices across the intercept-only, linear, quadratic, and conditional linear models; [Faubert, 2013](#)). The addition of this latent variable and the characterization demonstrated excellent fit indices; however, the addition of quadratic growth did not significantly improve the linear model $\Delta\chi^2(4) = 7.21, p = .125$. Thus, the linear model was the best characterization of the MOT performance trajectories.

A conditional linear model examined the presence of an ADHD, SLD, or IDD diagnosis (i.e., as separate predictor binary variables) while accounting for CPT-3 d' t -score, age, and sex. The model demonstrated excellent fit. The means and variances for both latent variables of the intercept and slope are reported in [supplemental Table 2](#). Additionally, the model revealed no statistically detectable covariance between the intercept and slope, suggesting that session 1 performance, or baseline MOT capability, did not predict MOT learning trajectory (see [supplemental table 2](#)). [Fig. 3](#) illustrates the learning trajectories for the entire sample and between the presence or absence of an ADHD, SLD, and IDD diagnosis. [Fig. 4](#) presents the conditional LGCM model with standardized and partially standardized estimates, standardized covariance, and residuals.

The intercept factor represents the intercept of the linear performance trajectory, and thus, is characterized as baseline performance on the MOT paradigm (session 1). Here, the presence of an IDD diagnosis resulted in a decrease in session 1 (baseline) performance on the MOT task compared to participants that did not have this diagnosis ($\hat{\gamma}_{\text{IDD-Int}} = -27.04, p < .001$). Additionally, age significantly predicted session 1 (baseline) performance on the MOT task ($\hat{\gamma}_{\text{Age-Int}} = -6.23, p < .001$), where performance increased with age. Furthermore, an ADHD ($\hat{\gamma}_{\text{ADHD-Int}} = 5.32, p = .404$), SLD diagnosis ($\hat{\gamma}_{\text{SLD-Int}} = -5.29, p = .533$), performance on the CPT-3 ($\hat{\gamma}_{\text{CPT-3-Int}} = -0.15, p = .649$), or Sex ($\hat{\gamma}_{\text{Sex-Int}} = 4.64, p = .467$) did not predict the intercept. Together, the predictors accounted for 22% of the variance in the intercept. These results suggest that there is a positive association between session 1 performance and age as well as session 1 performance and the presence of an IDD diagnosis. More specifically, MOT performance at baseline increased with age while individuals with an IDD diagnosis present diminished tracking capabilities compared to those that did not have an IDD diagnosis.

The slope factor measured the rate of change in MOT performance across 15 sessions, and thus, represents participants learning on the MOT task. The results demonstrated that the presence of ADHD was associated with a decrease in the slope ($\hat{\gamma}_{\text{ADHD-Slp}} = -0.80, p = .021$) compared to participants without an ADHD diagnosis. Similarly, the presence of SLD was associated with a decrease in the

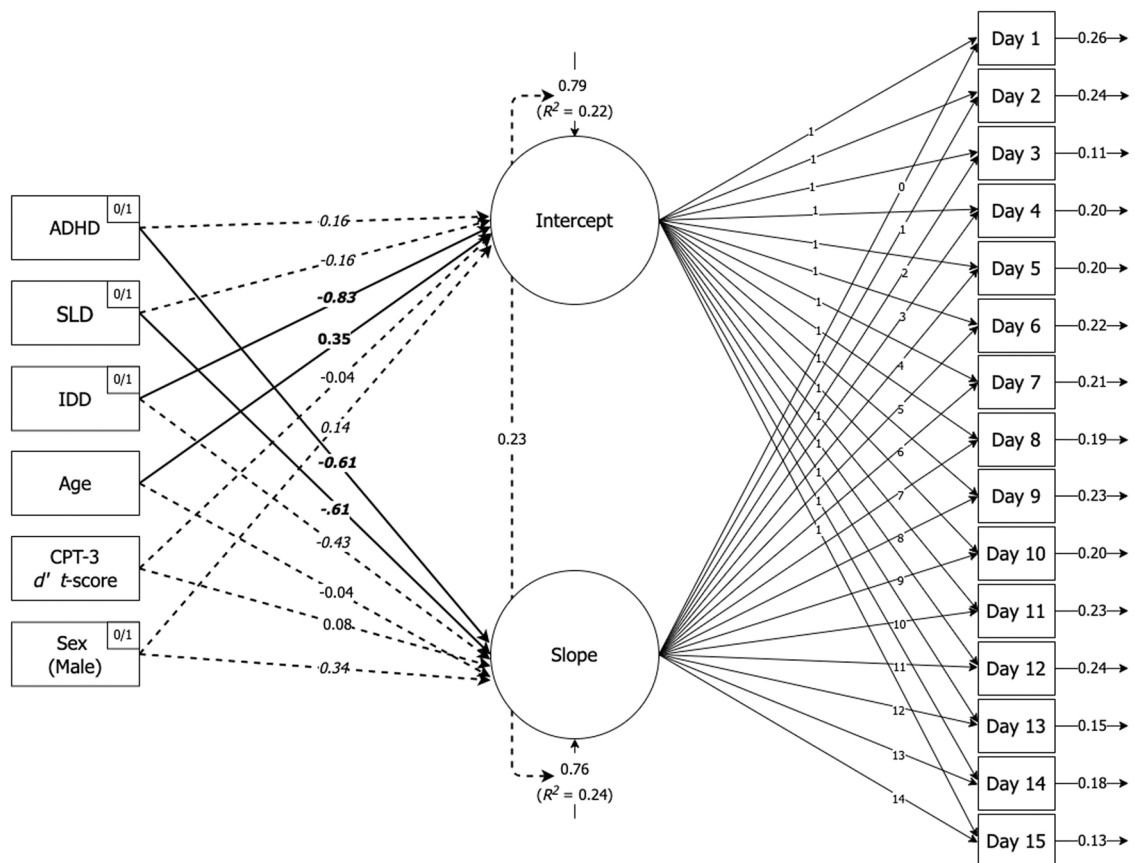


Fig. 4. Conditioned latent growth curve model for MOT performance trajectories. Standardized estimates, covariances and residuals are presented for the latent growth curve model. Partially standardized estimates for binary variables are italicized. Statistically detectable paths are represented by solid lines, and non-statistically detectable paths are represented by dashed lines.

slope ($\hat{\gamma}_{\text{SLD-Slp}} = -0.81, p = .032$). Neither CPT-3 performance ($\hat{\gamma}_{\text{CPT-Slp}} = 0.01, p = .493$), age ($\hat{\gamma}_{\text{Age-Slp}} = -0.03, p = .683$), the presence of an IDD diagnosis ($\hat{\gamma}_{\text{IDD-Slp}} = -0.57, p = .142$), nor sex ($\hat{\gamma}_{\text{Sex-Slp}} = .453, p = .155$) predicted the slope of MOT performance from session 1–15. Together, the predictors accounted for 24% of the variance in the slope. These results suggest that individuals with an ADHD or SLD diagnosis demonstrate attenuated learning of the MOT paradigm across 15 sessions compared to those that did not have an ADHD or SLD diagnosis, respectively.

4. Discussion

The aim of the current study was to characterize learning on an attention-based task as a function of diagnostic profiles that exhibit deficits in attention, learning, and intelligence (i.e., ADHD, SLD, and IDD, respectively). This was done by assessing participants' performance on an MOT task across 15 sessions, affording the opportunity to characterize learning trajectories across diagnostic profiles. MOT performance trajectories were analyzed using LGCM and a linear trajectory best represented the sample's performance. The linear trajectory suggests that overall, individuals with diverse neurodevelopmental conditions demonstrated learning on the task. Furthermore, the analysis attributed the variance in the performance trajectories as a function of the presence or absence of an ADHD, SLD, and IDD diagnosis. Given the current study's dichotomization of diagnostic profiles (i.e., the presence of the neurodevelopmental condition, compared to the absence of that neurodevelopmental condition), all findings should be interpreted using this relativistic reasoning. As shown in Fig. 3, a decreased baseline performance for participants with IDD was found, suggesting pre-existing, diminished attention resource capacities for individuals with an IDD diagnosis relative to a neurodivergent sample without this diagnosis. Therefore, using MOT task performance as a proxy measure of attention suggests that individuals with ADHD and SLD do not present pre-existing diminished attention resource capacities relative to other neurodevelopmental conditions. Examining the performance across multiple repeated sessions on the MOT task, that characterized the efficiency in learning on the paradigm, resulted in reduced slopes in the linear MOT performance trajectories for individuals diagnosed with ADHD and SLD.

Three distinct MOT performance trajectories exhibited across ADHD, SLD, and IDD diagnostic profiles characterize the learning capability on an attention-based measure. These findings and the approach (i.e., MOT) are clinically relevant in that an assessment of attention resource capacity and learning capability (i) are arguably more accessible and sensitive to attentional disturbances across different populations than current validated tasks (Meyerhoff & Papenmeier, 2020; Tullo, Faubert et al., 2018), and (ii) provide a more eclectic approach for understanding condition-specific strengths and deficits involved in attentional and learning capabilities.

In the present study, attention resource capacity was defined as the reserve of available resources for task demands. The significantly reduced MOT performance for IDD diagnostic profiles is congruent with the current theory linking MOT performance to attention resource capacity (Alnæs et al., 2014; Alvarez & Franconeri, 2007; Suchow et al., 2014; Tullo, Faubert et al., 2018). This result extends previous research characterizing individual differences in MOT capability in a typically-developing adult sample as a function of IQ (Oksama & Hyönä, 2004); that is, a proxy for higher-order cognition. More specifically, Tullo, Faubert et al. (2018) found a significant decreases in tracking capability for individuals 1 standard deviation below the population average in IQ. Here, marked differences were revealed across 1 standard deviation in IQ; however, the range in these differences was between 2 and 3 standard deviations below the population mean. The presence of an IDD diagnosis, including an IQ score within the vicinity of 70 (American Psychiatric Association, 2013), predicted decreased MOT baseline performance, further characterizing the impact of intelligence on attention resource capacity specific to atypically developing populations. Therefore, reduced attentional capability exhibited by IDD individuals at baseline (i.e., session 1) extends the characterization of individual differences in attention resource capacity with intelligence (i.e., or higher-order cognition) to children and adolescents with extremely low to borderline IQ.

Also, our results demonstrate that individuals with an IDD diagnosis required a significant number of sessions to reach baseline MOT performance exhibited by other diagnostic profiles (see Fig. 3). However, individuals with IDD did not demonstrate deficits in learning capability in visuo-attentional tracking given the rate of change in MOT performance across 15 sessions relative to other neurodevelopmental conditions. Therefore, it can be suggested that learning gains in IDD may be masked by diminished deficits in attention (i.e., as indicated by baseline performance; Deutsch et al., 2008), whereas the reduced rate of change in MOT performance trajectories for individuals with ADHD and SLD are suggestive of learning-deficits. These results are clinically relevant in that they suggest that IQ should be considered when interpreting measures of attention, particularly within the context of learning.

Our results revealed that individuals with ADHD and SLD demonstrated a reduced rate of linear change in task performance across the 15 sessions compared to those without ADHD and those without SLD, respectively. Both ADHD and SLD diagnostic profiles demonstrated reduced rates of change in MOT performance, our representation of learning capability, and this finding may at least in part reflect the high comorbidity between individuals diagnosed with ADHD and SLD (i.e., 45% as reported by DuPaul et al., 2013), which includes deficits in learning at a behavioural level (Carter et al., 1995; Hazell et al., 1999). In fact, individuals with ADHD and SLD present a vulnerability to distractors as demonstrated by reduced learning in a problem-solving task (Passolunghi et al., 2005). Previous research suggests individuals with ADHD present deficits in sustaining attentional resources to task demands while individuals with SLD present deficits in selecting relevant information from irrelevant information (Soo & Bailey, 2006). Unfortunately, our findings cannot support nor refute this discrepancy. Given the versatility of the MOT approach for isolating specific subcomponents of attention (Scholl, 2009), future research can explore the vulnerability to distractors by extending tracking duration (i.e., sustained; Tombu & Seiffert, 2008) or assigning an identity to each target item (i.e., selective; Oksama & Hyönä, 2016) within the MOT task.

4.1. Age, Sex, and CPT-3 as covariates

It is important to note that differences in baseline performance and learning capability on the MOT task between the presence of the

targeted neurodevelopmental conditions, were demonstrated while accounting for baseline attention ability using a separate and clinically validated measure of attention (i.e., CPT-3 performance), age, and sex. For instance, the results demonstrated that age was a statistically detectable predictor of the intercept; that is, participants' session 1 performance on the MOT task increased with age. This finding further extends previous results demonstrating a developmental trend in MOT capability (Brockhoff et al., 2016; Trick & Perl, 2005; Trick, Jaspers-Fayer et al., 2005), to individuals with a neurodevelopmental condition. However, age was not a statistically significant predictor of the slope in performance trajectories, which indicates that the task was learnable despite the recruited sample's significant age range. Given these findings, future research will be well served to further explore the development of MOT capability across typically and atypically developing populations in longitudinal designs. Conversely, sex was not a statistically detectable predictor of both session 1 performance and the rate of change on the MOT task suggesting that there are no sex differences in MOT capability, within a neurodivergent population.

Similarly, performance on a separate and clinically validated measure of attention, the CPT-3 was included in the LGCM. The results demonstrated that performance on the CPT-3 was not a statistically tenable predictor of baseline capability nor learning trajectories on the MOT task. While it is surprising that a clinically validated measure of attention did not predict MOT capability or improvement in MOT capability given that tracking is a robust indicator of attention (Huang et al., 2012), we discuss possible explanations for this occurrence. For one, the interpretation of this finding is that CPT-3 performance was not a statistically detectable predictor of baseline performance or learning trajectories on the MOT above and beyond the diagnostic profiles, age, sex, and IQ predictors. As such, it could be that the variance was better explained by the other predictors. Secondly, the sample recruited for the study, that is, individuals attending schools that provide specialized services for students with alternative learning profiles may have reduced the variance in CPT-3 performance. In fact, the mean score on the clinically validated measure was a full standard deviation above the population mean and the sample variance was below the population variance (see Table 1; Conners, 2014). Collectively, the sample's performance on the CPT-3 task reached below-average to problematic levels of attention (see Fig. 2B). Lastly, there were no differences in performance on the task between the presence and absence of the three neurodevelopmental conditions suggesting reduced performance on this measure across diagnostic profiles of interest. Taken together, the inclusion of CPT-3, sex, and age as covariates strengthens the interpretation and generalizability of the diagnostic predictors given that individual differences factors beyond diagnostic profiles were accounted for in the models.

4.2. Limitations

While the current study demonstrated distinct differences in learning trajectories for participants with an ADHD and SLD diagnosis, a possible limitation is highlighted by the inability to examine the impact of comorbidity between ADHD, SLD, and IDD. We attribute this limitation to a lack of statistical power to add additional predictors, such as the impact of comorbid diagnoses. Similarly, the sample size for those diagnosed with SLD was significantly smaller than the other diagnostics groups, and therefore, these results should be interpreted accordingly. Instead, this study serves as a benchmark for future studies to explore the interaction between deficits in attention and learning to further characterize the impact of deficits in attention and/or learning in learning an attention-based task. Additionally, the overlap between ADHD and SLD was significant; and thus, a direct comparison between the two neurodevelopmental conditions is limited. However, the dichotomous structure of the SLD predictor compares individuals with an SLD diagnosis to individuals diagnosed with any other neurodevelopmental condition, which demonstrates decreased learning for individuals with SLD compared to their peers. Moreover, future research can expand on these differences between the targeted neurodevelopmental conditions with the inclusion of neurotypical participants without deficits in attention, learning, or intelligence to further qualify the impact of these deficits in learning an attention-based task. Given the current study's limitation in dichotomizing diagnostic status, the inclusion of a neurotypically developing group would attenuate the differences in baseline and rate of change on the MOT task between the neurodevelopmental conditions of interest. As such, the exclusion of this group may have afforded the opportunity to explore differences in attention resource capacity and learning mechanisms within a neurodivergent population in a way not otherwise possible. Nevertheless, the similarity in performance for individuals expressing deficits in attention or learning highlights the need to assess learning when diagnosing ADHD and attention when diagnosing SLD (DuPaul et al., 2013; Mayes et al., 2000; Zablotzky et al., 2019). This bidirectional consideration can be informative throughout assessment, diagnosis, and treatment recommendations.

4.3. Alternative use for MOT

The observations on reduced attention resource capacity for IDD and the difficulties with learning across repeated practice sessions for ADHD and SLD diagnostic profiles contribute to the validity of an approach utilizing MOT to characterize attention (Huang et al., 2012; Tullo, Faubert et al., 2018). These findings extend previous research characterizing differences between neurodevelopmental conditions, where differences in tracking capability were investigated by and attributed to limitations in visual cognition rather than attention (Evers et al., 2014; O'Hearn et al., 2010; Van der Hallen et al., 2015). Instead, MOT can isolate the allocation of attentional resources to relevant and irrelevant information (Alvarez & Franconeri, 2007; Tullo, Faubert et al., 2018).

In addition to characterizing differences in learning trajectories across deficits in attention, learning, and intelligence, previous research has explored the feasibility of improving attention with MOT (Parsons et al., 2014; Thompson et al., 2016; Tullo & Guy, 2018; Vartanian et al., 2016). Specific to individuals with neurodevelopmental conditions, our results demonstrate a distinct increase in MOT performance suggestive of learning, which replicates previous findings (Tullo & Guy, 2018). Similarly, there was no relationship between baseline performance and learning trajectory. Together, these findings provide additional evidence that MOT is learnable and

therefore suitable for individuals with below-average to borderline intelligence and with problematic levels of attention. The application of LGCM has yielded an opportunity to describe learning trajectories in MOT and the results from the current study highlight the potential to better understand engagement with and progression through cognitive-based interventions in an effort to improve efficacy (Orylska et al., 2019).

5. Conclusion

The results of the current study characterize condition-specific deficits as a function of learning an attention-based paradigm, where individuals with IDD demonstrate reduced attentional capability and individuals with ADHD and SLD present difficulties in learning. Coupling the robust measure of attention with an advanced statistical methodology, such as LGCM, has resulted in the characterization of learning an attention-based task, specific to a portion of the population that is typically underserved (Tullo & Guy, 2018). The present study highlights the role of higher-order cognition (i.e., intelligence) in attention resource capacity and demonstrates the negative impact deficits in attention or learning have in regard to repeated practice on MOT. Overall, these results (i) suggest consideration of attention, learning capability, and intelligence when diagnosing either ADHD, SLD, or ID; and (ii) further advocate for MOT as a descriptor of attention.

Funding

This work was supported by the Canadian Institutes of Health Research (CIHR) Postdoctoral Fellowship (#176576; Canada), the Fonds de recherche du Québec – Santé (FRQS) Postdoctoral Fellowship (#305855; Canada) awarded to DT and a William Dawson Scholar (Canada) awarded to AB.

CRedit authorship contribution statement

Domenico Tullo: Conceptualization, Methodology, Investigation, Project administration, Formal analysis, Writing – original draft, Visualization. **Jocelyn Faubert:** Software, Writing – review & editing. **Armando Bertone:** Supervision, Resources, Writing – review & editing.

Competing interests

JF is the director of Faubert Lab at the University of Montreal and he is the inventor of the commercial version of the Multiple Object Tracking task used in this study, NeuroTracker. In this capacity, he holds shares in the company. AB is a scientific advisor for NeuroTracker, the commercial version of the Multiple Object Tracking task used in this study.

Data Availability

The authors do not have permission to share data.

Acknowledgements

We would like to thank the Summit Center for Education, Research, and Training (SCERT) for their collaboration and for facilitating this research project.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ridd.2022.104402](https://doi.org/10.1016/j.ridd.2022.104402).

References

- Alnæs, D., Sneve, M. H., Espeseth, T., Endestad, T., van de Pavert, S. H. P., & Laeng, B. (2014). Pupil size signals mental effort deployed during multiple object tracking and predicts brain activity in the dorsal attention network and the locus coeruleus. *Journal of Vision*, 14(4), 1–20. <https://doi.org/10.1167/14.4.1>
- Alvarez, G. A., & Franconeri, S. L. (2007). How many objects can you track?: Evidence for a resource-limited attentive tracking mechanism. *Journal of Vision*, 7(13), 1–10. <https://doi.org/10.1167/7.13.14>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders: DSM-5*. (5th ed.). American Psychiatric Association.
- Baker, B. L., Neece, C. L., Fenning, R. M., Crnic, K. A., & Blacher, J. (2010). Mental disorders in five-year-old children with or without developmental delay: Focus on ADHD. *Journal of Clinical Child & Adolescent Psychology*, 39(4), 492–505. <https://doi.org/10.1080/15374416.2010.486321>
- Bauer, D.J., & Curran, P.J. (2019). *Structural equation modeling: R demonstration notes (Version 2019.3)*. Curran-Bauer Analytics. (<https://curranbauer.org/wp-content/uploads/2019/04/SEM-R-notes-2019-3.pdf>).
- Boyle, C. A., Boulet, S., Schieve, L. A., Cohen, R. A., Blumberg, S. J., Yeargin-Allsopp, M., Visser, S., & Kogan, M. D. (2011). Trends in the prevalence of developmental disabilities in US children, 1997–2008. *Pediatrics*, 127(6), 1034–1042. <https://doi.org/10.1542/peds.2010-2989>

- Brockhoff, A., Papenmeier, F., Wolf, K., Pfeiffer, T., Jahn, G., & Huff, M. (2016). Viewpoint matters: Exploring the involvement of reference frames in multiple object tracking from a developmental perspective. *Cognitive Development*, 37, 1–8. <https://doi.org/10.1016/j.cogdev.2015.10.004>
- Carter, C. S., Krenner, P., Chaderjian, M., Northcutt, C., & Wolfe, V. (1995). Abnormal processing of irrelevant information in attention deficit hyperactivity disorder. *Psychiatry Research*, 56(1), 59–70. [https://doi.org/10.1016/0165-1781\(94\)02509-H](https://doi.org/10.1016/0165-1781(94)02509-H)
- Case, R. (1985). *Intellectual development: Birth to adulthood*. Academic Press.
- Conners, K.C. (2014). *Conners continuous performance Test 3rd edition (Conners CPT 3) & Conners Continuous Auditory Test of Attention (Conners CATA): Technical Manual*. Multi-Health Systems Inc.
- Deutsch, C. K., Dube, W. V., & McLivane, W. J. (2008). Attention deficits, Attention-Deficit Hyperactivity Disorder, and intellectual disabilities. *Developmental Disabilities Research Reviews*, 14(4), 285–292. <https://doi.org/10.1002/ddrr.42>
- Doshier, B. A., Han, S., & Lu, Z.-L. (2010). Information-limited parallel processing in difficult heterogeneous covert visual search. *Journal of Experimental Psychology: Human Perception and Performance*, 36(5), 1128–1144. <https://doi.org/10.1037/a0020366>
- DuPaul, G. J., Gormley, M. J., & Laracy, S. D. (2013). Comorbidity of LD and ADHD: Implications of DSM-5 for assessment and treatment. *Journal of Learning Disabilities*, 46(1), 43–51. <https://doi.org/10.1177/0022219412464351>
- Engle, R. W. (2002). Working memory capacity as executive attention. *Current Directions in Psychological Science*, 11(1), 19–23.
- Engle, R. W., Kane, M. J., & Tuholski, S. W. (1999). Individual differences in working memory capacity and what they tell us about controlled attention, general fluid intelligence, and functions of the prefrontal cortex. In A. Miyake, & P. Shah (Eds.), *Models of working memory: Mechanisms of active maintenance and executive control*. Cambridge University Press.
- Enneser, F., & Medioni, G. (1995). Finding Waldo, or focus of attention using local color information. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(8), 805–809. <https://doi.org/10.1109/34.400571>
- Evers, K., de-Wit, L., Hallen, R. V., der, Haesen, B., Steyaert, J., Noens, I., & Wagemans, J. (2014). Brief report: Reduced grouping interference in children with ASD: evidence from a multiple object tracking task. *Journal of Autism and Developmental Disorders*, 44(7), 1779–1787. <https://doi.org/10.1007/s10803-013-2031-4>
- Faubert, J. (2013). Professional athletes have extraordinary skills for rapidly learning complex and neutral dynamic visual scenes. *Scientific Reports*, 3, 1–3. <https://doi.org/10.1038/srep01154>
- Faubert, J., & Sidebottom, L. (2012). Perceptual-cognitive training of athletes. *Journal of Clinical Sport Psychology*, 6(1), 85.
- Fombonne, E., Zakarian, R., Bennett, A., Meng, L., & McLean-Heywood, D. (2006). Pervasive developmental disorders in Montreal, Quebec, Canada: Prevalence and links with immunizations. *Pediatrics*, 118(1), e139–e150. <https://doi.org/10.1542/peds.2005-2993>
- Forster, S., & Lavie, N. (2008). Failures to ignore entirely irrelevant distractors: The role of load. *Journal of Experimental Psychology: Applied*, 14(1), 73.
- Forster, S., Robertson, D. J., Jennings, A., Asherson, P., & Lavie, N. (2014). Plugging the attention deficit: Perceptual load counters increased distraction in ADHD. *Neuropsychology*, 28(1), 91.
- Gathercole, S. E., & Alloway, T. P. (2008). *Working memory and learning: A practical guide for teachers*. Sage Publications.
- Government of Québec. (2013). Organization of educational services for at-risk students and students with handicaps. *Social Maladjustments Oregon Learning Difficulties*. (<http://www.education.gouv.qc.ca/en/references/tx-solrtperecherhepublicationtx-solrpublicationnouveau/results/detail/>).
- Hazell, P. L., Carr, V. J., Lewin, T. J., Dewis, S. A. M., Heathcote, D. M., & Brucki, B. M. (1999). Effortful and automatic information processing in boys with ADHD and Specific Learning Disorders. *The Journal of Child Psychology and Psychiatry and Allied Disciplines*, 40(2), 275–286. <https://doi.org/10.1111/1469-7610.00441>
- Huang, L., Mo, L., & Li, Y. (2012). Measuring the interrelations among multiple paradigms of visual attention: An individual differences approach. *Journal of Experimental Psychology: Human Perception and Performance*, 38(2), 414–428. <https://doi.org/10.1037/a0026314>
- Kane, M. J., Brown, L. H., McVay, J. C., Silvia, P. J., Myin-Germeys, I., & Kwapil, T. R. (2007). For whom the mind wanders, and when: An experience-sampling study of working memory and executive control in daily life. *Psychological Science*, 18(7), 614–621. <https://doi.org/10.1111/j.1467-9280.2007.01948.x>
- Kelley, T. A., & Yantis, S. (2009). Learning to attend: Effects of practice on information selection. *Journal of Vision*, 9(7). <https://doi.org/10.1167/9.7.16>, 16–16.
- Klingberg, T. (2013). *The learning brain: Memory and brain development in children*. Oxford University Press.
- Legault, L., & Faubert, J. (2012). Perceptual-cognitive training improves biological motion perception: Evidence for transferability of training in healthy aging. *Neuroreport*, 23(8), 469–473.
- Levitt, H. (2005). Transformed up-down methods in psychoacoustics. *The Journal of the Acoustical Society of America*, 49(2B), 467–477. <https://doi.org/10.1121/1.1912375>
- Makovski, T., & Jiang, Y. V. (2009). The role of visual working memory in attentive tracking of unique objects. *Journal of Experimental Psychology: Human Perception and Performance*, 35(6), 1687–1697. <https://doi.org/10.1037/a0016453>
- Mayes, S. D., Calhoun, S. L., & Crowell, E. W. (2000). Learning disabilities and ADHD overlapping spectrum disorders. *Journal of Learning Disabilities*, 33(5), 417–424. <https://doi.org/10.1177/002221940003300502>
- Melby-Lervåg, M., Redick, T. S., & Hulme, C. (2016). Working memory training does not improve performance on measures of intelligence or other measures of “far transfer”: Evidence from a meta-analytic review. *Perspectives on Psychological Science: A Journal of the Association for Psychological Science*, 11(4), 512–534. <https://doi.org/10.1177/1745691616635612>
- Meyerhoff, H. S., & Papenmeier, F. (2020). Individual differences in visual attention: A short, reliable, open-source, and multilingual test of multiple object tracking in PsychoPy. *Behavior Research Methods*. <https://doi.org/10.3758/s13428-020-01413-4>
- O’Hearn, K., Hoffman, J. E., & Landau, B. (2010). Developmental profiles for multiple object tracking and spatial memory: Typically developing preschoolers and people with Williams syndrome. *Developmental Science*, 13(3), 430–440. <https://doi.org/10.1111/j.1467-7687.2009.00893.x>
- Oksama, L., & Hyönä, J. (2004). Is multiple object tracking carried out automatically by an early vision mechanism independent of higher-order cognition? An individual difference approach. *Visual Cognition*, 11(5), 631–671. <https://doi.org/10.1080/13506280344000473>
- Oksama, L., & Hyönä, J. (2016). Position tracking and identity tracking are separate systems: Evidence from eye movements. *Cognition*, 146, 393–409. <https://doi.org/10.1016/j.cognition.2015.10.016>
- Orylska, A., Hadwin, J. A., Kroemeke, A., & Sonuga-Barke, E. (2019). A growth mixture modeling study of learning trajectories in an extended computerized working memory training programme developed for young children diagnosed with Attention-Deficit/Hyperactivity Disorder. *Frontiers in Education*, 4. <https://doi.org/10.3389/feduc.2019.00012>
- Parsons, B., & Faubert, J. (2021). Enhancing learning in a perceptual-cognitive training paradigm using EEG-neurofeedback. *Scientific Reports*, 11(1), 4061. <https://doi.org/10.1038/s41598-021-83456-x>
- Parsons, B., Magill, T., Boucher, A., Zhang, M., Zogbo, K., Bérubé, S., Scheffer, O., Beauregard, M., & Faubert, J. (2014). Enhancing cognitive function using perceptual-cognitive training. *Clinical EEG and Neuroscience*, 47(1), 37–47. <https://doi.org/10.1177/1550059414563746>
- Passolunghi, M. C., Marzocchi, G. M., & Fiorillo, F. (2005). Selective effect of inhibition of literal or numerical irrelevant information in children with Attention Deficit Hyperactivity Disorder (ADHD) or Arithmetic Learning Disorder (ALD). *Developmental Neuropsychology*, 28(3), 731–753. https://doi.org/10.1207/s15326942dn2803_1
- Pondé, M. P., Cruz-Freire, A. C., & Silveira, A. A. (2011). Relationship between learning problems and attention deficit in childhood. *Journal of Attention Disorders*, 16(6), 505–509. <https://doi.org/10.1177/1087054711423625>
- Powell, B. J., Beidas, R. S., Lewis, C. C., Aarons, G. A., McMillen, J. C., Proctor, E. K., & Mandell, D. S. (2017). Methods to improve the selection and tailoring of implementation strategies. *The Journal of Behavioral Health Services & Research*, 44(2), 177–194. <https://doi.org/10.1007/s11414-015-9475-6>
- Pylshyn, Z. W., & Storm, R. W. (1988). Tracking multiple independent targets: Evidence for a parallel tracking mechanism. *Spatial Vision*, 3(3), 179–197. <https://doi.org/10.1163/156856888X00122>
- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(1), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Serif, G. (2010). Attention trajectories, mechanisms and outcomes: At the interface between developing cognition and environment. *Developmental Science*, 13(6), 805–812. <https://doi.org/10.1111/j.1467-7687.2010.01013.x>

- Scholl, B. J. (2009). What have we learned about attention from multiple-object tracking (and vice versa). In *Computation, cognition and pylyshyn* (pp. 49–77). MIT Press.
- Seitz, A. R., & Watanabe, T. (2005). A unified model for perceptual learning. *Trends in Cognitive Sciences*, 9(7), 329–334. <https://doi.org/10.1016/j.tics.2005.05.010>
- Seitz, A. R., & Watanabe, T. (2009). The phenomenon of task-irrelevant perceptual learning. *Vision Research*, 49(21), 2604–2610. <https://doi.org/10.1016/j.visres.2009.08.003>
- Soo, C. A., & Bailey, J. G. (2006). A review of functioning of attentional components in children with Attention-Deficit/Hyperactivity Disorder and Learning Disabilities. *Brain Impairment*, 7(2), 133–147. <https://doi.org/10.1375/brim.7.2.133>
- Steinmayr, R., Ziegler, M., & Träuble, B. (2010). Do intelligence and sustained attention interact in predicting academic achievement. *Learning and Individual Differences*, 20(1), 14–18. <https://doi.org/10.1016/j.lindif.2009.10.009>
- Strong, R. W., & Alvarez, G. A. (2017). Training enhances attentional expertise, but not attentional capacity: Evidence from content-specific training benefits. *Journal of Vision*, 17(4). <https://doi.org/10.1167/17.4.4>, 4–4.
- Suchow, J. W., Fougner, D., Brady, T. F., & Alvarez, G. A. (2014). Terms of the debate on the format and structure of visual memory. *Attention, Perception, & Psychophysics*, 76(7), 2071–2079. <https://doi.org/10.3758/s13414-014-0690-7>
- Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285.
- Sweller, J. (1994). Cognitive load theory, learning difficulty, and instructional design. *Learning and Instruction*, 4(4), 295–312. [https://doi.org/10.1016/0959-4752\(94\)90003-5](https://doi.org/10.1016/0959-4752(94)90003-5)
- Thompson, T. W., Waskom, M. L., & Gabrieli, J. D. E. (2016). Intensive working memory training produces functional changes in large-scale frontoparietal networks. *Journal of Cognitive Neuroscience*, 28(4), 575–588. https://doi.org/10.1162/jocn_a.00916
- Tomblu, M., & Seiffert, A. E. (2008). Attentional costs in multiple-object tracking. *Cognition*, 108(1), 1–25. <https://doi.org/10.1016/j.cognition.2007.12.014>
- Trick, L. M., Jaspers-Fayer, F., & Sethi, N. (2005). Multiple-object tracking in children: The “Catch the Spies” task. *Cognitive Development*, 20(3), 373–387. <https://doi.org/10.1016/j.cogdev.2005.05.009>
- Trick, L. M., Perl, T., & Sethi, N. (2005). Age-related differences in multiple-object tracking. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 60(2), P102–P105. <https://doi.org/10.1093/geronb/60.2.P102>
- Tullo, D., Faubert, J., & Bertone, A. (2018). The characterization of attention resource capacity and its relationship with fluid reasoning intelligence: A multiple object tracking study. *Intelligence*, 69, 158–168. <https://doi.org/10.1016/j.intell.2018.06.001>
- Tullo, D., Guy, J., Faubert, J., & Bertone, A. (2018). Training with a three-dimensional multiple object-tracking (3D-MOT) paradigm improves attention in students with a neurodevelopmental condition: A randomized controlled trial. *Developmental Science*, 21(6), Article e12670. <https://doi.org/10.1111/desc.12670>
- Tullo, D., Perico, C., Faubert, J., & Bertone, A. (2020). Multiple object-tracking isolates feedback-specific load in attention and learning. *Journal of Vision*, 20(5), 1–15. <https://doi.org/10.1167/jov.20.5.9>
- Van der Hallen, R., Evers, K., de-Wit, L., Steyaert, J., Noens, I., & Wagemans, J. (2015). Multiple object tracking reveals object-based grouping interference in children with ASD. *Journal of Autism and Developmental Disorders*, 1–9. <https://doi.org/10.1007/s10803-015-2463-0>
- Vartanian, O., Coady, L., & Blackler, K. (2016). 3D multiple object tracking boosts working memory span: Implications for cognitive training in military populations. *Military Psychology*, 28(5), 353–360. <https://doi.org/10.1037/mil0000125>
- Wechsler, D. (2011). *Wechsler abbreviated scale of intelligence—Second Edition (WASI-II)*. San Antonio, TX: NCS Pearson.
- Wilson, B. J., Will, M. N., Schoenfeld-McNeill, J., & Montague, R. (2013). The attention skills of boys with and without developmental delays: Associations with academic performance. *Early Child Development and Care*, 183(5), 651–666. <https://doi.org/10.1080/03004430.2012.682727>
- Zablotsky, B., Black, L. I., Maenner, M. J., Schieve, M. J., Danielson, M. L., Bitsko, R. H., Blumberg, S. J., Kogan, M. D., & Boyle, C. A. (2019). Prevalence and trends of developmental disabilities among children in the United States: 2009–2017. *Pediatrics*, 144(4). <https://doi.org/10.1542/peds.2019-0811>
- Zeithamova, D., & Maddox, W. T. (2006). Dual-task interference in perceptual category learning. *Memory & Cognition*, 34(2), 387–398. <https://doi.org/10.3758/BF03193416>